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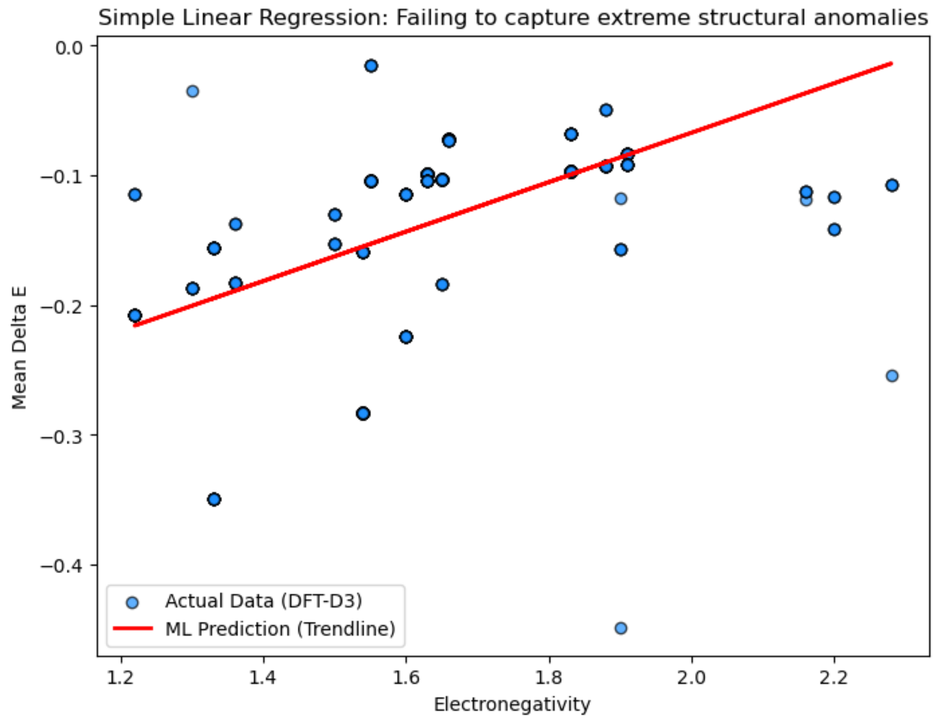
# Building a Regression Model

# A Simple Linear Regression Model

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--- SIMPLE LINEAR REGRESSION RESULTS ---
R-squared (R²): 0.2247
Mean Squared Error (MSE): 0.0040

Interpretation: An R² score close to 0 proves that electronegativity ALONE
cannot reliably predict the extreme DFT-D3 breakdowns.
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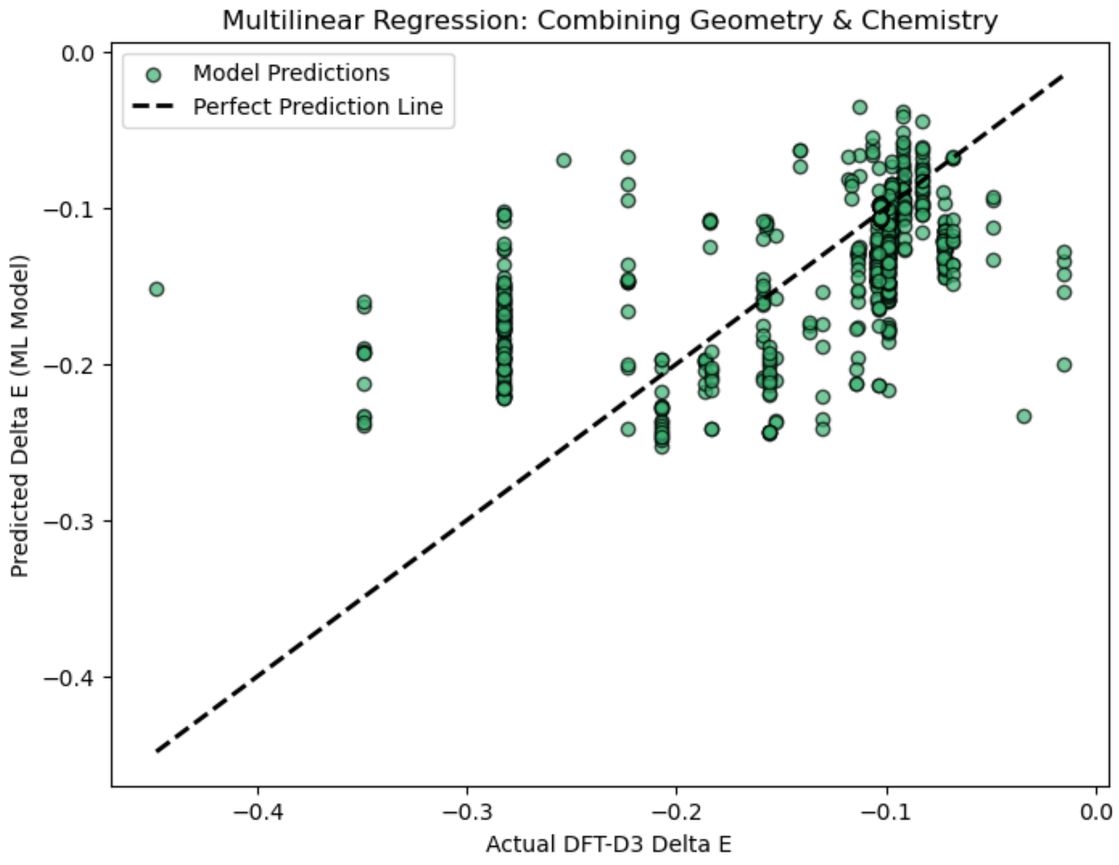
# Multilinear Regression Model

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--- MULTILINEAR REGRESSION RESULTS ---
New R-squared (R²): 0.3626
New Mean Squared Error (MSE): 0.0032

--- THE PREDICTIVE EQUATION ---
Delta E = (0.1209 * Electronegativity) + (-0.1527 * Packing) + -0.2586
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- The Limitation of Chemistry Alone: A Simple Linear Regression using only electronegativity yielded a poor correlation ($R^2 = 0.22$), proving that chemical identity alone cannot predict extreme dispersion breakdowns.

- Multilinear Improvement: By combining both chemical (electronegativity) and geometric (packing fraction) features, the model's predictive power significantly improved.

- The Predictive Equation: $\Delta E = 0.121(\text{Electronegativity}) - 0.153(\text{Packing}) - 0.259$

- Physical Interpretation: The model mathematically confirms our earlier observations. Higher electronegativity (more covalent) and tighter lattice packing both push towards zero, meaning the DFT-D3 method is more robust for dense, late d-block oxides.